

A HOLISTIC ASSESSMENT OF **TOYOTA'S SUPPLY CHAIN--** SYNERGY OF STRATEGY, COLLABORATION, TECHNOLOGY, RISK AND SUSTAINABILITY

MGMT 90026 Supply Chain Management
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Total Word Account: 4879

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Introduction

This report analyses Toyota's global supply chain strategy, focusing on its ethical and sustainability practices. It covers the core dimensions of production model design, strategic alliance building, digital transformation, supply chain resilience building and ESG performance. Through lean production, stable supplier partnerships and the application of advanced technologies, Toyota has achieved significant results in cost efficiency and operational agility. However, it still faces challenges in Scope 3 emissions control, low-tier supplier compliance, and electrification transformation.

In this regard, the report recommends that Toyota further promote the use of blockchain technology for supply chain traceability management and increase support for supplier renewable energy transition to improve overall transparency, reduce environmental impacts, and strengthen its position as a global leader in sustainable manufacturing. The Overarching Supply Chain Strategy of Toyota

Toyota's supply chain strategy, a benchmark in the global manufacturing industry, has been based on "Lean Production" for a considerable period (Sugimori et al., 1977). This strategy emphasizes core principles such as Just-in-Time, zero inventory, pull management and Kaizen. As the global environment becomes increasingly complex, Toyota's strategy has undergone a gradual shift from a purely efficiency-oriented approach to an enhanced resilience and agile response, forming a comprehensive supply chain system that integrates cost control, personalized services and risk prevention. This section will analyze its core strategic modules, including pull strategy, Postponement strategy, MTO and MTS hybrid model and enhanced network structure.

1. The Overarching Supply Chain Strategy of Toyota

1.1 Pull strategy and Kanban system

The Pull System is the foundation of Toyota's supply chain management. Its core concept is to trigger production and replenishment based on actual customer demand. The system uses the Kanban System to synchronize information between processes and control material flow. Sugimori et al. (1977) define JIT as a production method that is synchronized with the actual needs of customers, producing only the required quantity when needed, in order to reduce inventory and waste. Moreover, Kanban is a visual material and information transfer tool used to control the flow between processes (Adnan et al., 2013). Kanban is divided into "pickup cards" and "production cards" to indicate downstream pull and upstream production, ensuring that each process only performs operations according to demand, thereby significantly reducing inventory, avoiding overproduction, and alleviating the "bullwhip effect" (Huang et al., 2014). Toyota has expanded the Kanban system to its global parts supply network, establishing a highly controllable production rhythm management mechanism through standardized cards, visual management and information sharing. Brintrup et al. (2015) point out that this demand-driven information feedback mechanism not only improves production efficiency but also enhances the rapid recovery capability of the entire system to emergencies, such as natural disasters, which is the key foundation for its "agility".

1.2 Postponement and Modular Design

To cope with high variability and global diversified market demands, Toyota widely applies the strategy of manufacturing postponement in its production system. Fisher (1997) emphasizes that, based on standard module prefabrication, the final customization and assembly, such as interior, paint, configuration, will be delayed until the order is confirmed, so as to balance economies of scale and personalized response.

This strategy is especially appropriate for product lines with high demand uncertainty, such as hybrid models and electric vehicles, and functions in conjunction with Toyota's modular platform architecture. Toyota also employs postponement logistics to achieve localized configuration of cross-regional inventory, thereby enhancing overall flexibility from the end of

the supply chain (Sugimori, 1977). Brintrup et al. (2015) suppose that this strategy enhances the "adaptability" of the supply chain and constitutes a pivotal element of Toyota's resilience strategy.

1.3 Hybrid MTS and MTO Strategy

Toyota adopts a hybrid production model that combines "Make-to-Order (MTO)" and "Make-to-Stock (MTS)" according to product types and market characteristics. MTS is based on forecasts to achieve economies of scale, while MTO starts production only after the customer confirms the order to achieve personalized customization. The MTS strategy is adopted on standardized main models, like Corolla, to improve capacity utilization and delivery efficiency, while the MTO strategy is adopted on high-end or customized models, like Lexus, to meet personalized needs (Lane, 2020).

This flexible hybrid model enhances Toyota's resource coordination capabilities between different markets. Yang (2022) points out that Toyota integrates ERP systems and order data to achieve intelligent switching and plan matching between MTS and MTO, which not only meets service levels but also controls inventory risks, forming flexible manufacturing advantages in a high-mix-low-batch environment.

1.4 Network Structure and Strategic Evolution

In recent years, Toyota has undergone a significant shift in its strategic approach, responding to global challenges such as the pandemic, earthquakes, and chip shortages. The automotive manufacturer has transitioned from a primarily "cost-efficiency-oriented" to a "network resilience-oriented" model. Kito et al. (2014) conducted a network structure analysis of Toyota's supply chain, which revealed that it possesses a small-world structure (Small-World Network) and multi-center redundancy. In the event of the failure of some nodes, the system is capable of maintaining operation through alternative paths.

Lee (2004) proposes that this structure provides real-world case support for the Triple-A mode. Toyota has effectively enhanced the resilience and sustainable competitiveness of its global supply chain in an uncertain environment by developing three key capabilities: The qualities of agility, adaptability, and alignment are of particular significance in this context.

2. Strategic Alliances of Toyota

2.1 Toyota's Strategic Alliances

The success of Toyota's supply chain is contingent not only on internal efficiency, but also on the strategic alliance model that the company employs with external partners, characterized by high levels of collaboration and long-term trust. Cox (1999) demonstrates that Toyota rejects traditional transactional cooperation and instead builds close cooperative relationships with core suppliers and logistics partners through the concept of "co-evolution" to achieve resource sharing, capability building and risk sharing.

2.2 Keiretsu: Embedded Supplier Networks

Toyota has established a deep-rooted cooperative mechanism with key suppliers (e.g. Denso and Aisin) through the "Keiretsu" system, encompassing cross-equity, joint R&D, and shared training. The system has been shown to enhance information transparency, relationship stability, and collaborative efficiency for both parties (Kito et al., 2014). Moreover, Toyota has promoted horizontal information sharing and improvement mechanisms by establishing a supplier association "Kyohokai" (Brintrup et al., 2011). The association provides suppliers with regular reviews and a co-creation learning platform, with the objective that suppliers not only maintain consistency with Toyota, but also cooperate with each other, thereby building a collaborative network based on trust.

2.3 Strategic Collaboration with Logistics Providers

In terms of logistics, Toyota has established a deep cooperative relationship with Toyota Tsusho (a logistics subsidiary) and third-party logistics service providers (such as NYK Logistics). The scope of this cooperative endeavor encompasses not only the execution of transportation logistics and the administration of warehousing facilities, but also extends to the domains of order forecasting, delivery scheduling, and the integration of information technology systems (Yang, 2022).

This phenomenon can be interpreted as an evolution in the nature of the supply chain 3PL relationship, transitioning from a model of "service outsourcing" to a more collaborative approach of "process co-construction". Toyota has implemented a series of measures to address

this issue, including the introduction of delayed delivery solutions in numerous regions. These solutions involve deferring the configuration of terminal vehicles until the sales market is complete, with the aim of enhancing local response speed and reducing global inventory levels.

2.4 Coordination Mechanisms and Information Sharing

To enhance alliance collaboration, Toyota has deployed ERP and EDI systems extensively, with the objective of visualizing the complete information flow from order entry to delivery. In addition, Brown (2005) states that Toyota and its suppliers have adopted the CPFR (Collaborative Planning, Forecasting and Replenishment) mechanism to facilitate the joint formulation of production and sales plans, with the objective of ensuring the optimal alignment between inventory and delivery capabilities.

Attaran and Attaran (2007) state that these technologies and process means have been shown to improve the efficiency of cross-enterprise collaboration, as well as deepening the level of supplier participation in strategy formulation and execution. Furthermore, they have been demonstrated to improve the consistency and system execution of the supply chain (Attaran & Attaran, 2007).

To sum up, Toyota builds its global supply chain system with lean logic, showing strong resilience and high responsiveness in an uncertain environment. It is based on pull production, combined with delay strategy and flexible hybrid mode to achieve a balance between cost control and service agility. By building the Keiretsu supply network, integrating third-party logistics and ERP/EDI systems, Toyota has created a highly collaborative and visual global operations platform. This model not only improves operational efficiency and customer responsiveness but also enhances self-recovery in the face of risk, making it a model for the manufacturing industry to cope with complex environments and achieve sustainable competitiveness.

3. Technologies of Toyota's Supply Chain

Toyota's global supply chain is supported by advanced technologies to enhance visibility, agility, efficiency, and sustainability. These technologies include Electronic Data Interchange (EDI), digital twin systems, cloud computing, robotics, artificial intelligence (AI), Internet of

Things (IoT). Each one is important for the performance of the supply chain.

3.1 Internet of Things (IoT)

Internet of Things (IoT) technologies, including RFID sensors, provide Toyota with end-to-end supply chain visibility. RFID is often considered a prerequisite for the Internet of Things, as it identifies objects, records, or controls targets through radio waves or computers (Jia, 2012). IoT enables real-time tracking of inventory, equipment, and vehicles across production and distribution networks. This is helpful for Toyota's just in time production model, as the accuracy of material delivery time. The Internet of Things can also detect potential faults and arrange maintenance in advance by collecting device operation data. It has become a new trend for unified tracking, monitoring, management, and processing of the entire manufacturing process and after-sales management of automobiles (Liu, 2012).

3.2 Artificial Intelligence (AI)

Artificial Intelligence (AI) used to analyze market trends, historical data, and external variables to predict demand and optimize inventory planning, making the final decision for the supply chain. The goal is to mitigate the bullwhip effect and reduce instances of stockouts or overproduction. Artificial intelligence helps enhance competitive advantage in optimizing automotive supply chain planning and decision-making (Kumar, 2024). Jones (2025) further explains that AI enhances the precision of forecasting models and contributes to more efficient procurement, capacity planning, and risk management practices across global supply chains.

3.3 Robotics & Automation

Toyota's assembly lines and logistics centers are using **robots and automation**. For example, Automated Forklift, Automated Guided Vehicles. These technologies have improved work efficiency, reduced human errors, and maintained the same level of operational quality. Reduce Toyota's dependence on manual labor. Toyota's forklift division reports that self-driving carts move materials efficiently inside factories, improving safety and reducing the need for manual labor. Research by Sahoo's team (2022) states that intelligent manufacturing builds efficient smart production systems that can adapt fast to market changes or worker shortages.

3.4 Digital twins

Toyota uses digital twins to create virtual models of its supply chain, so it can test different situations like sudden increases in demand change or problems from suppliers. According to Tao et al. (2018), digital twins help companies make better decisions by predicting what might happen and making decisions by supporting data. It can help Toyota see how its supply chain is running, improve arrangement of inventory, and check the delivery process. Scenario planning for interruptions, such as natural disasters or geopolitical issues. Optimization of inventory, warehousing, and distribution. In addition, Boschert and Rosen (2016) confirmed that digital twins improve traceability and enable positive responses to system failures or inefficiencies, enhancing the adaptability of the supply chain.

3.5 Cloud computing

Cloud computing and big data analysis are the core technologies of Toyota's digital transformation. The cloud provides flexible systems for storing data and accessing it in real time, while big data analytics helps Toyota pull useful insights from massive amounts of supply chain data. These tools mainly aim to connect data across different regions, make demand forecasts more precise, and speed up decision-making. For example, cloud platforms connect factories, logistics partners, and suppliers to a shared data hub. Research by Waller and Fawcett (2013) shows that big data analytics improves supply chain visibility by spotting patterns or unusual data points, which helps plan logistics better. Similarly, Choi (2018) found that cloud computing cuts IT equipment costs and makes it easier for different levels of the supply chain to work together.

3.6 Electronic Data Interchange (EDI)

Electronic Data Interchange (EDI) is a key technology that enables Toyota and its suppliers to transfer business data quickly. By using standardized formats and encrypted transmission, it ensures both data security and standardized operational processes. EDI works effectively in handling repetitive daily tasks automatically, like processing orders, creating invoices, and sending shipping updates. Its main goals are to cut down on mistakes made by people, make data more reliable, and help businesses run more smoothly. In Toyota's production system, working closely with suppliers is key. EDI keeps communication in sync and supports just-in-

time production. For example, Zhang and colleagues (2011) found that EDI helps share data across different parts of the supply chain, making partnerships with suppliers stronger. Lee's team (2003) also showed that EDI shortens the time needed for orders, lowers costs, and makes the entire supply chain simpler and more efficient. Toyota's 'Just-in-time' (JIT) production model enables smooth information flow synchronization.

3.7 Recommendations for technologies

3.7.1 3D Printing

3D printing can help Toyota to further enhance the responsiveness of its supply chain. The spread application of 3D printing will lead to manufacturing flowing towards countries closer to end customers. Manufacturers can produce components on demand nearby, reducing dependence on distant suppliers and minimizing delivery time and cost (Chen,2016). In the context of supply chain management, 3D printing can support the availability of spare parts, achieve mass customization, and reduce inventory holding costs (Rogers, 2016). 3D printing, also known as Additive Manufacturing (AM), is a technology that surpasses traditional production and makes mechanical component production more accessible. In the future, 3D printers will penetrate the automotive industry (Mohanavel, 2022). Furthermore, Kubáč (2017) explains that 3D printing enhances supply chain agility by allowing firms to respond to design changes or demand fluctuations more flexibly. For Toyota, adopting 3D printing would align well with its idea and Environmental Challenge 2050 by reducing material waste and emissions from transportation.

3.7.2 Autonomous Supply Chain Control Towers

Toyota to incorporate Autonomous Supply Chain Control Towers, powered by AI and advanced analytics. These digital platforms provide end-to-end visibility, intelligent alerting, and autonomous decision-making capabilities. A control tower could proactively monitor global supply disruptions, suggest alternative logistics routes, and balance inventory across regions. Automated decision-making, optimizing algorithms to balance resources in real-time (such as production capacity, transportation vehicles, warehouse space), maximizing utilization, and reducing costs. However, to significantly

enhance responsiveness during crises, as global supply chains become more unstable, these systems would allow Toyota to maintain resilience while reducing reliance on human intervention (Bennett, 2023).

3.8 Interpretation of Contribution Scores in the Chart

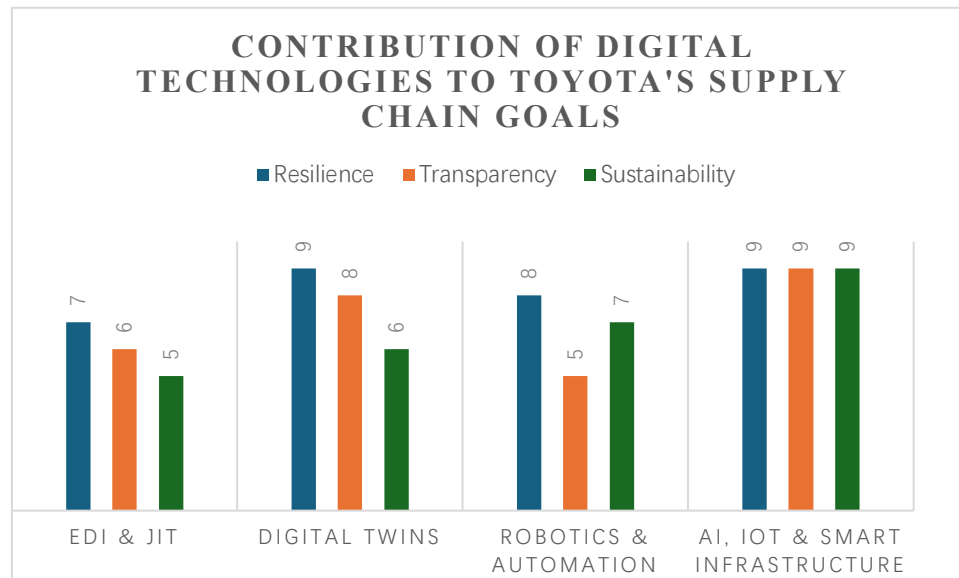


Chart scores are based on academic literature and actual Toyota cases, reflecting the extent to which each technology supports the three goals of resilience, transparency, and sustainability (on a scale of 1-10).

3.8.1 EDI and JIT: Improve process efficiency through real-time synchronisation and lean operations, with moderate performance on resilience and transparency (Zhang et al., 2016; Lee et al., 2022), but less impact on sustainability.

3.8.2 Digital twins: with prediction and scenario simulation capabilities, enhanced risk response and visualisation capabilities, and high resilience scores (Tao et al., 2019).

3.8.3 Robotics and automation: Improves operational stability and energy use efficiency, contributing more to sustainability but with slightly less visibility (Srai et al., 2021).

3.8.4 AI, IoT and Intelligent Systems: fully support the three main objectives to improve forecasting accuracy, chain-wide tracking and energy optimisation (Chopra & Meindl, 2023; Simchi-Levi et al., 2023).

4. Toyota Global Supply Chain Risk Analysis and Response Strategies

Toyota is a leading enterprise in the global automotive industry. However, in recent years, the company has faced various risks, such as frequent supply chain disruptions, multiple recall incidents caused by product quality issues, and persistent compliance risks. All these have had a negative impact on the company's reputation and financial situation. Next, I will conduct an in-depth analysis of the main risk factors of enterprises in their global operations, focus on studying the weak links existing in supply chain management, and accordingly propose two sets of targeted improvement plans to enhance Toyota's risk prevention and response capabilities.

4.1 Risk Identification: Three Core Risks

I have summarized the three main risks that Toyota is currently facing. First of all, there is the complexity of the supply chain. It is well known that although Toyota emphasizes its advantages of minimizing inventory and synchronizing production processes with its renowned Just-in-Time (JIT) production system. However, this efficiency-driven model is usually limited by complex and diverse automotive parts suppliers. For instance, if a single automotive parts supplier fails to deliver components within the agreed time, it may lead to the shutdown of the entire automotive production line. Furthermore, although Toyota's global supply chain partnerships can reduce costs, they will also become increasingly vulnerable to unexpected events such as natural disasters, epidemics or logistics failures (Choi, Rogers, & Vakil, 2020). The second risk is quality control and recalls. Toyota has experienced multiple recall issues over the past 20 years. Among them, the recall of over 9 million vehicles due to product quality accidents from 2009 to 2021 has drawn much attention. Recently, in 2020, Toyota recalled over 3.4 million vehicles due to faults in the electronic control unit (ECU) that affected the operation of airbags in cars (Toyota, 2020). These incidents not only exposed the systemic problems of Toyota's lack of quality control and supplier supervision, but also the recall of cars would have impacts such as financial costs and brand reputation on the company. The third risk is geopolitics and global economic stability. For instance, events such as covid-19 and the Russia-

Ukraine conflict that occurred in recent years have affected Toyota's global supply chain system. In addition, economic recession, inflation and currency fluctuations have further complicated the global supply chain, which has greatly affected Toyota's management issues and costs in the global supply chain.

4.2 Response strategy: Current measures in place

To alleviate and address the current risks, Toyota has implemented several strategies to reduce the risks in its global supply chain. These strategies include lean production, diversified alternative suppliers, and a sound quality management system. First, Toyota's lean production model aims to improve efficiency and reduce waste, and its core is the "Just-In-Time" (JIT for short) production method. This mode requires that components be delivered to the assembly line only when needed, thereby minimizing inventory occupation and costs (Ohno, 1988). However, JIT also has certain risks, especially when the supply chain is disrupted by unexpected events. For instance, after the great earthquake in northeastern Japan in 2011, some Toyota factories were forced to suspend production because key parts could not be delivered on time. Such experiences prompted Toyota to make adjustments to its original strategy. On the one hand, the JIT principle was still adhered to. But on the other hand, for key components, Toyota began to increase safety stock and reevaluate the inventory strategy (Sheffi, 2007). Secondly, regarding the supply chain, Toyota originally adopted a relatively concentrated strategy, establishing close ties with a few long-term cooperative suppliers. Although this strategy is conducive to controlling quality and reducing transaction costs, it may also increase systemic risks in some cases. For instance, after the earthquake in 2011, the production suppliers of some key parts of Toyota lost their supply capacity, resulting in production disruptions. To this end, Toyota has gradually promoted a supplier diversification strategy, especially by increasing alternative suppliers for key components. Meanwhile, the company also advocates local procurement and establishes local supply chains in different countries or regions to reduce transportation costs and cross-border risks (Nair, Narasimhan, & Bendoly, 2011). Finally, the reason why Toyota has long been regarded as a model of quality management is largely attributed to its systematic quality control system. From the production line to supplier management, Toyota has implemented the concept of Total Quality

Management (TQM). Toyota also conducts regular quality reviews of suppliers to ensure that their production standards meet the company's requirements. From 2009 to 2010, Toyota experienced a large-scale vehicle recall in North America, raising questions from the outside world about its quality control. After this incident, Toyota increased its investment in problem tracking, data analysis, and recall response mechanisms. Meanwhile, it reunified the quality standards and sense of responsibility of its Global subsidiaries through "Toyota Global Vision".

4.3 Build a digital risk control system

I suggest that Toyota should establish a dynamic supply chain risk management system based on digital technology. At present, the supply chain is confronted with multiple challenges such as natural disasters and geopolitical conflicts. Relying solely on traditional response methods is no longer sufficient. It is suggested to advance in three steps: Firstly, deploy Internet of Things sensors at key nodes to monitor the production and transportation status of components in real time; Secondly, utilize AI to analyze historical data and predict potential risk points; Finally, establish an emergency response mechanism to automatically trigger the plan when the system issues a warning. For example, an intelligent dispatching platform can be developed to automatically switch logistics routes when ports are congested. If this system had been in place during the earthquake in 2011, the alternative solutions could have been activated at least 48 hours in advance. Meanwhile, it is suggested to share the data platform with the main suppliers to achieve traceability of quality information. This can not only ensure the JIT efficiency but also improve the speed of abnormal response. This digital transformation needs to be implemented in phases, with priority given to covering the supply chain of core components such as engines.

4.4 Improve supply chain resilience

Secondly, I suggest that Toyota should enhance the resilience of its supply chain to reduce the risks of recall incidents and global supply disruptions. This can be achieved by increasing the diversity of suppliers and enhancing the transparency of each link in the supply chain. For instance, establish a regional supplier system, increase the procurement channels for key components, and reduce reliance on a single supplier, thereby enhancing flexibility and adaptability. Meanwhile, Toyota can introduce advanced technologies such as the Internet of

Things (IoT) and blockchain to conduct real-time monitoring of the production and logistics transportation processes of key automotive parts. These digital tools not only help to quickly identify potential defects but also narrow the recall range and reduce their impact when problems occur. By building a more sensitive and transparent supply chain, Toyota will be able to identify and respond to potential risks more proactively, rather than merely passively responding to the occurrence of problems.

5. A Study of Toyota's Ethical and Sustainable Supply Chain Strategy

5.1 Sustainable Sourcing and Supplier Collaboration

Toyota has taken the initiative to focus on sustainable sourcing and has developed close relationships with suppliers. Sustainable Purchasing Guidelines indicate that the company must comply with environmental, social, and legal standards across its supply network (Salminen & Laari, 2024). In the pre-contract dialogues, Toyota encourages suppliers to adopt such practices through mutual trust and long-term partnerships. Toyota is also part of Drive Sustainability, a platform of 11 other automotive companies that promote sustainability in their supply chains (Palhau et al., 2024). With this initiative, Toyota develops solutions to address suppliers' challenges such as carbon neutrality, sustainable raw materials, workforce well-being, and circular value chains. Through this collaborative approach, Toyota helps suppliers obtain training and compliance tools and encourages them to improve their sustainability performance (Palhau et al., 2024).

5.2 Carbon Neutrality and Environmental Impact Reduction

Toyota's supply chain strategy is based on its long-term goal of carbon neutrality by 2050 and the Toyota Environmental Challenge 2050. The company is reducing greenhouse gas emissions across the entire vehicle lifecycle, from raw material extraction to manufacturing and end-of-life recycling (Toyota Corporation, 2021). To meet Toyota's carbon reduction goals, the company has set clear environmental criteria that suppliers must meet through its Green Supplier Requirements. For example, Toyota has recycling systems, such as using recycled

PET bottles in the vehicle interior (Toyota Corporation, 2021). Additionally, Toyota includes renewable energy solutions such as its investment in EODev (a hydrogen energy company) to decarbonize its operations.

5.3 Ethical Labor Practices and Human Rights

Toyota pays excellent attention to ethical labor practices and human rights within the supply chain. The company's Supplier CSR Guidelines 2021 cover key issues such as forced labor, child labor, and fair wages. In addition, the company has regular supplier surveys and on-the-ground audits to ensure its guidelines are followed (Wada, 2021). Toyota is firmly committed to ethical procurement and sources Fairtrade-certified cotton for uniforms. It also commits to sustainable palm oil procurement by joining the Roundtable on Sustainable Palm Oil (Toyota Tshusho Corporation, 2018). These initiatives specifically aim to protect workers' rights and enhance the working conditions in developing countries known for labor rights abuse.

5.4 Supply Chain Resilience and Localization

Toyota's sustainability efforts depend on the supply chain resilience. Toyota localizes manufacturing plants and diversifies its supplier base to diminish the environmental impact of transportation and lead times. Toyota has optimized its logistics by, for example, using rail freight to transport vehicles between France and the UK, reducing CO2 emissions in the process, in Europe (Williams, 2022). As such, Toyota's production model is based on what is known as its Just-in-Time (JIT) system, ensuring that supply chain disruptions are minimized by maintaining lean inventories and flexible supplier relationships. This way, Toyota has been able to avoid and minimize many supplies chain related issues. The system was tried in the 2011 Fukushima disaster, where Toyota quickly reorganized its sourcing strategy to ensure continuity while keeping up with its ethical and environmental standards.

5.5 Technology and Transparency

Toyota incorporates advanced technology to enhance the efficiency and sustainability of its supply chain through its integration with technology. For example, AI-based design processes reduce waste as they minimize design iterations, while digital logistics solutions, such as planning of vehicles and reduction of emissions. As part of the current trends in the traceable and ethically sourced goods industry, Toyota is yet another company that uses blockchain

technology to increase supply chain transparency (Toyota Corporation, 2021). Through sharing sustainability data with its partners, Toyota collaboration with other players to deal with emissions, waste, and labor issues is evident. It is doing this by making sure its direct and indirect suppliers adhere to sustainability standards.

5.6 Critical Analysis of Toyota's Approach

Toyota's supply chain strategy is a comprehensive commitment to the sustainability of the environment and social responsibility. However, challenges remain; although supplier self-reporting can be relied upon, it is more uncertain among lower-tier suppliers due to the lack of resources to implement sustainability measures. Additionally, while Toyota has reduced its Scope 1 and 2 emissions, Scope 3 emissions, which represent 70 percent of corporate emissions, are a difficult target to achieve as the company's global supply chain is complex (Mukunde, 2021). Furthermore, critics have targeted Toyota's conservative approach towards fully electric vehicles (EVs), preferring hybrid alternatives, in an era when competitors increasingly devote themselves to electric mobility.

5.7 Recommendations for Improvement

5.7.1 Implement Blockchain-Based Supply Chain Traceability

To boost the transparency and accountability of its supply chain, Toyota ought to implement blockchain technology, which would track materials and parts through its supply chain. Blockchain's immutable ledger system promises to give assurance to the source of materials such as natural rubber, cobalt, and lithium, which are required for electric vehicles, coupled with ethical issues, into the system. This technology would give Toyota real-time tracking capabilities and enable the company to enforce its Sustainable Purchasing Guidelines better and respond much faster to noncompliance issues (Moosavi et al., 2021). Combining blockchain would further Toyota's efforts to do business ethically and place the company at the leading edge of supply chain transparency, an important attribute for the clean-conscious consumer and investor.

5.7.2 Increasing the Progress of Transition to Renewable Energy across Suppliers

Toyota should further support suppliers in moving to renewable energy sources, building on its

existing initiatives. Toyota could adopt a model similar to Apple's Supplier Clean Energy Program by providing financial and technical assistance to its suppliers to install solar or wind energy systems (Lin & Lu, 2024). The move would significantly cut Scope 3 emissions from suppliers, which Toyota needs to achieve carbon neutrality. Suppose Toyota's suppliers must set specific renewable energy targets, such as 50% renewable energy usage by 2030. In that case, Toyota can make a huge cut in its supply chain's carbon footprint and bypass the regulatory risks imposed by strict environmental laws on key markets.

Conclusion

Toyota has built a global supply chain system centred on lean and transparency, demonstrating its long-term commitment to sustainability and ethical responsibility. Despite positive progress, challenges remain in Scope 3 emissions, lower tier supplier compliance and electrification transition. In the face of an increasingly stringent global regulatory environment, Toyota needs to accelerate its blockchain traceability system, promote the use of clean energy, and strengthen its risk response capabilities in order to strengthen its position as a leader in sustainable manufacturing.

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Appendix

Record of group meetings

Date	3 . 27
Participants	Danqi Liu ; Haofu Deng; Xuanzhen Wu Yufei Liu
Purpose of the meeting	Conduct research on potential topics.
Content	According to the topic selection preferences of each team member, search for reference materials on alternative topics related to the supply chain in accordance with the report requirements.

Date	4 . 8
Participants	Danqi Liu ; Haofu Deng; Xuanzhen Wu Yufei Liu; Zhifeng Song
Purpose of the meeting	Determine the topic. Analyze and divide the report structure.
Content	The report is divided into the following parts: Yufei Liu: Toyota's overall supply chain strategy and its changes over time, and the nature of its strategic alliances. Haofu Deng: Supply chain management technology

	portfolio, effectiveness evaluation and technology optimization suggestions. Xuanzhen Wu: Risk challenges and response strategies and optimization suggestions in global operations. Danqi Liu: Current strategies and optimization suggestions for building ethical and sustainable supply chains. Zhifeng Song: Introduction, Toyota supply chain digital empowerment, summary, and full text structure content verification.
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Date	4 . 15
Participants	Danqi Liu ; Haofu Deng; Xuanzhen Wu Yufei Liu; Zhifeng Song
Purpose of the meeting	Summarize the difficult points in each part.
Content	Difficulty points: The first part of the report lacks material on the overall strategy and the nature of the alliance, and requires a large number of reference materials to provide support. At the same time, this part should echo the 6 recommendations that appear later.

Date	4 . 20
Participants	Danqi Liu ; Haofu Deng; Xuanzhen Wu Yufei Liu; Zhifeng Song
Purpose of the meeting	Restate and integrate the suggestions in the three main paragraphs.
Content	Re-watch the video from week 3, review the environmental issues that have arisen during Toyota's development, and adjust the content of the "Risk challenges" section of the report. Change the focus of sustainable development to product recalls and environmental damage.

Date	4 . 26
Participants	Danqi Liu ; Haofu Deng; Xuanzhen Wu Yufei Liu; Zhifeng Song
Purpose of the meeting	Review the entire report, delete the cumbersome sentences, and modify the introduction according to the body paragraph.
Content	After integrating the last 6 suggestions and checking for duplicate or redundant content, we made deletions and changes. We also made slight adjustments to the report's opening chapter, "Overall Strategic Narrative and the

	Nature of Its Alliances”.
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Date	5 . 3
Participants	Danqi Liu ; Haofu Deng; Xuanzhen Wu Yufei Liu; Zhifeng Song
Purpose of the meeting	Review according to report requirements.
Content	Arrange the report structure, title, check references, and make a table of contents.